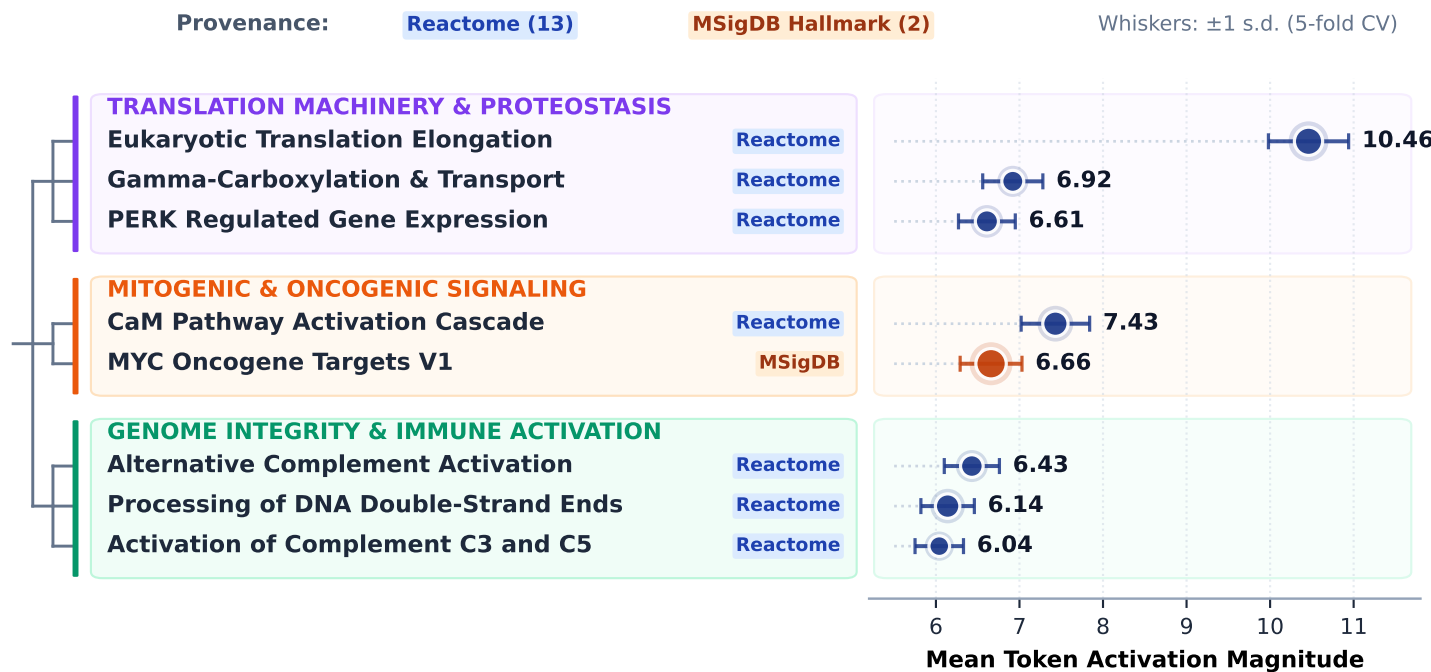
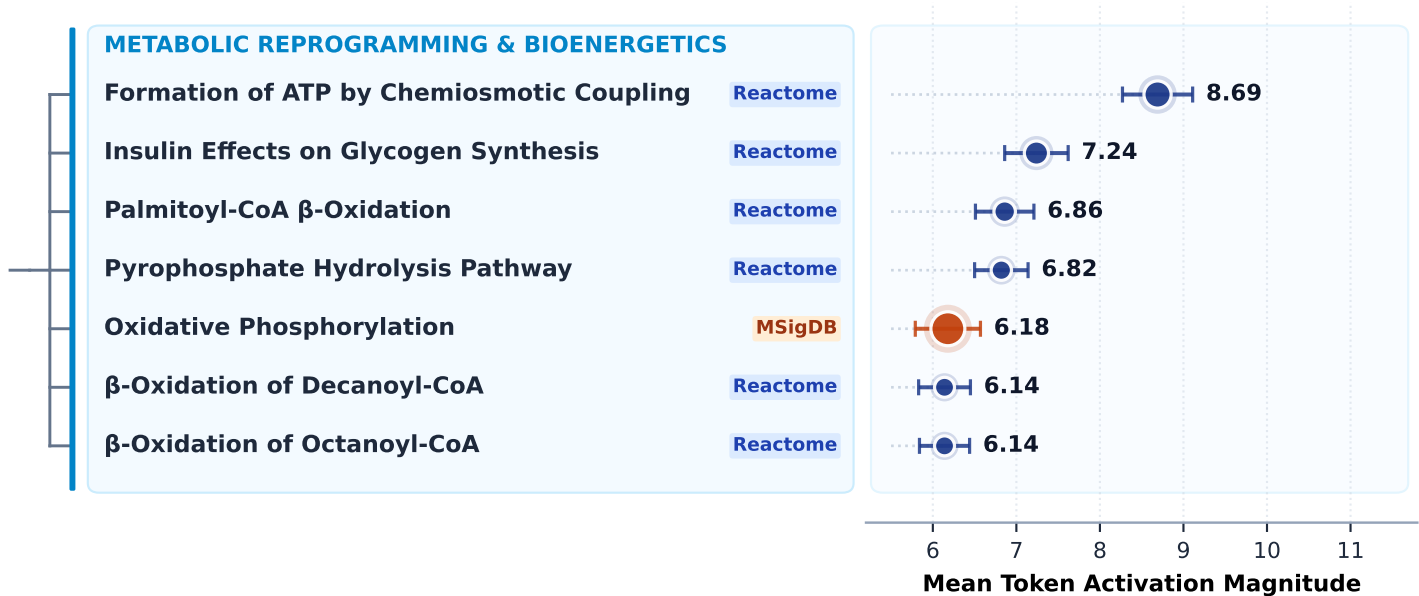
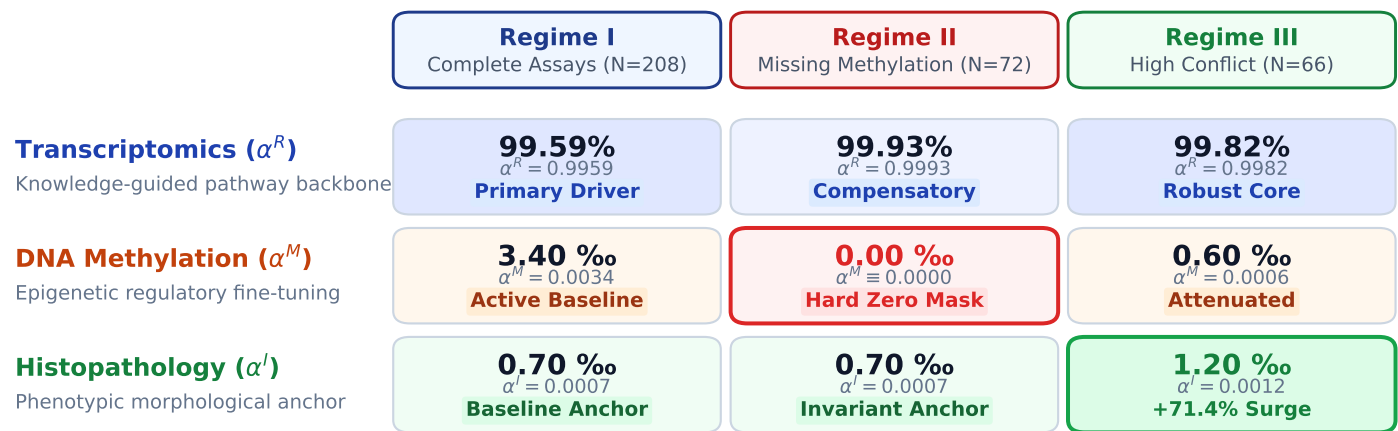


a Hierarchical pathway cladogram and biological token activations across cancer hallmarks



b Modality attention allocation and mechanism matrix across clinical regimes



c Dynamic state-transition trajectories across clinical regimes

